Ali Mamaghani — Curriculum Vitae

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Education

University of California San Diego **La Jolla, CA, USA**
Ph.D. in Electrical and Computer Engineering *2024-2030 (expected)*
GPA-**3.9/4**
Advisor: Prof. Dinesh Bharadia

Sharif University of Technology **Tehran, Iran**
Master's in Electrical Engineering (Communication Systems) *2022-2024*
GPA-**19.10/20.00(4/4)**
Thesis: Applications of LLMs in a Cloud-Native 5G Platform
Advisors: Prof. Babak Hossein Khalaj and Prof. Navid Nikaein (EURECOM)

Sharif University of Technology **Tehran, Iran**
Bachelor's in Electrical Engineering (Communication Systems) *2018-2022*
GPA-**19.15/20.00(3.99/4)**
Advisor: Prof. Babak Hossein Khalaj

Allameh Helli High School (NODET) **Tehran, Iran**
Mathematics and Physics Diploma. *2014-2018*
NODET: National Organization for Development of Exceptional Talents

Young Scholars Club **Tehran, Iran**
IPhO Team Selection Camp *Jun 2017- Sep 2017*

Research Interests

- Communication Networks
- Large Language Models
- Machine Learning
- Reinforcement Learning
- LLM Agents
- ML for Networking

Research Experience & Publications

Question Answering over Telecom Log Systems *Jan 2025 - Present*
Research Project

- *Supervisors:* Prof. Dinesh Bharadia (UC San Diego) and Prof. Srinivas Shakkottai (Texas A&M)
- Proposed LENS, a log-evidence-grounded neuro-symbolic agent for natural-language analytics over 5G traces, built on the premise that million-line logs should never be handed to an LLM directly: an offline pass decomposes raw logs into structured events in DuckDB and derives an empirical finite-state model of recurring transitions into a Trace Analytics Ontology, so that at query time the model synthesizes a structurally validated, query-specific analytics program executed as SQL — resolving the event pairing, identifier joins, grouping and arithmetic that attention alone gets wrong, and keeping sensitive traces on operator-controlled near-edge infrastructure where smaller open-weight models can run.
- *Publications:*
  - A. Mamaghani, Q. Zheng, U. Ghosh, V. Rajagopalan, S. Shakkottai, et al., "Log Pruning for Efficient Q&A in 5G Networks," *Proc. 2nd ACM Workshop on Open and AI RAN*, pp. 36–42, 2025.
  - O. Kamatani, S. Saruwatari, S. Seshasayee, A. Mamaghani, D. Bharadia, "LLM-5GMAC: Performance Optimization in O-RAN Split 7.2 Using LLM-Based MAC-Layer Log Analysis," *Proc. 2nd ACM Workshop on Open and AI RAN*, pp. 1–7, 2025.
  - Q. Zheng, U. Ghosh, A. Mamaghani, S. Shakkottai, D. Bharadia, "PHOENIX: Protocol State Machine Recovery from Events in 5G Logs," *Proc. 16th ACM Workshop on Wireless of the Students, by the Students, and for the Students (S3)*, 2025.

Benchmarking LLM Software-Engineering Agents on Telecom Codebases *2026*
Research Project

- *Collaborators:* Prof. Vijay K. Shah (NC State) and Prof. Dinesh Bharadia (UC San Diego)
- Built TeleSWEBench, the first commit-driven benchmark for whether coding agents can modify a mathematically rigorous wireless stack: real srsRAN developer commits distilled into 734 test cases with executable unit tests across three difficulty tiers, paired with TeleJudge, a hierarchical LLM-as-a-Judge that scores per file and aggregates holistically where unit tests cannot reach — exposing weak fault localization in AIDER, OpenHands and Claude Code, whose best configurations ship only ∼25% of changes.
- *Publication:*

- P. Gajjar, A. Mamaghani, D. Bharadia, V. K. Shah, "TeleSWEBench: A Commit-Driven Benchmark for Evaluating LLM-Powered Software Engineering in Telecommunications," *Proc. Annual Conf. of the Asia-Pacific Chapter of the Assoc. for Computational Linguistics (AACL-IJCNLP)*, Main Conference, 2026.

5GArena: LLM Agents that Build and Run Real 5G Experiments *Oct 2025 - Present*

Research Project

- *Supervisors:* Prof. Dinesh Bharadia (UC San Diego) and Prof. Srinivas Shakkottai (Texas A&M)
- Built 5GArena, an "experiments-as-code" harness in which every component (core, gNB, UE, RIC) is a declaratively-constructed runner driven over a lightweight JSON-RPC control plane instead of SSH glue (0.14 ms loopback round-trip, peak 11.2k calls/s; a per-host agent reads the RAN host's gNB metrics 28× faster than tunneling the same read over SSH), letting an LLM agent autonomously build and run real over-the-air 5G experiments on a live band-n78 testbed (open5gs core, srsRAN gNB on a USRP N310, EdgeRIC real-time RIC, COTS handsets) through a manifest-driven surface of 31 tools: running an experiment is a natural-language prompt and adding a capability is one runner method plus one manifest line.
- *Publication:*
  - A. Mamaghani, Q. Zheng, S. Saruwatari, S. Seshasayee, S. G. Shakkottai, D. Bharadia, "5GArena: A Harness for LLM Agents that Build and Run Real 5G Experiments on a Live O-RAN Testbed," *IEEE Int. Conf. on Communications (ICC)*, 2027. (in preparation)

Joint RAN and GPU-Compute Scheduling for Robot and LLM Tenants *Aug 2026 - Present*

Research Project

- *Supervisor:* Prof. Dinesh Bharadia (UC San Diego)
- Treated the radio and the edge GPU as one control problem for tenants that share both — a robot policy, uplink video and a handset LLM chat client on a live srsRAN/Open5GS n78 cell — building an EdgeRIC muApp that fuses RAN telemetry with GPU job state, attributes the bottleneck to either domain in common units, and actuates in both; this located control authority precisely: the GPU policy dominates (40–70× on the robot) while the RAN uplink weight is structurally a cap, not a floor, confirmed from scheduler source, measurement, online Q-learning and a control-mode sweep.

Tiny-Twin: CPU-Native Full-Stack Digital Twin for NextG *Jan 2025 - Present*

Research Project

- *Supervisors:* Prof. Dinesh Bharadia (UC San Diego), Prof. Srinivas Shakkottai (Texas A&M), and Prof. Ish Jain (RPI)
- Built Tiny-Twin, a CPU-native full-stack digital twin that occupies the missing middle between low-fidelity link-level simulators and high-cost FPGA/GPU emulators such as Colosseum: it couples time-varying multi-tap channel convolution — computed entirely in software on commodity CPUs via a redesigned architecture and system-level optimizations — with a complete 5G protocol stack, so multipath fading, Doppler spread and time-frequency selectivity actually reach MCS selection, beamforming and scheduling instead of being abstracted away into an averaged SNR or CQI trace. Built-in real-time RIC integration and per-UE channel isolation scale it to multiple concurrent UEs while preserving protocol timing and end-to-end behaviour, making AI-driven RAN algorithms and protocol designs testable reproducibly without radios or spectrum; released open source with plug-and-play replay of channel traces.
- *Publications:*
  - A. Mamaghani, U. Ghosh, S. Shakkottai, D. Bharadia, I. K. Jain, "Tiny-Twin: A CPU-Native Full-stack Digital Twin for NextG Cellular Networks," *IEEE Int. Symp. on Spectrum Innovation (DySPAN)*, pp. 1–8, 2026.
  - A. Mamaghani, U. Ghosh, S. Shakkottai, D. Bharadia, I. K. Jain, "Demo: Tiny-Twin – A CPU-Native Full-Stack Digital Twin for NextG Cellular Networks," *IEEE Int. Symp. on Spectrum Innovation (DySPAN)*, 2026.

EdgeRIC for OAI *Sep 2024 - Jan 2025*

Research Project

- *Supervisors:* Prof. Dinesh Bharadia (UC San Diego) and Prof. Srinivas Shakkottai (Texas A&M)
- Brought reinforcement learning into the RAN's real-time control loop: integrated EdgeRIC, a real-time radio intelligence controller, into the OpenAirInterface 5G RAN so that RL policies act on fine-grained, per-UE telemetry within the scheduling deadline, raising user throughput and making the RAN's allocation decisions application-aware rather than traffic-agnostic.

Applications of LLMs in a Cloud-Native 5G Platform from EURECOM *Apr 2023 - Aug 2024*

Master's Thesis Project

- *Supervisors:* Prof. Babak Hossein Khalaj and Prof. Navid Nikaein (EURECOM)
- Asked what a language model can do for an operator once the network itself is software: built a cloud-native 5G network on Kubernetes from OAI components, then layered LLM capabilities across it — explaining test output and errors in plain language, correcting configuration files through a GPT-3.5-turbo feedback loop, fine-tuning GPT-3.5-turbo to generate configurations, and building an assistant chatbot over the platform.
- *Publications:*

- A. Mamaghani, A. Nourian, N. Mohtaram, A. Shokrani, S. M. Nasiri, et al., "LLM for 5G: Network Management," *IEEE Int. Conf. on Machine Learning for Communication and Networking (ICMLCN)*, 2024.
- A. Majlesara, A. Majlesi, A. Mamaghani, A. Shokrani, B. H. Khalaj, "5G Network Automation Using Local Large Language Models and Retrieval-Augmented Generation," *arXiv:2511.21084*, 2025.

Implementation of 4G and 5G Mobile Networks with a Centralized Controller *Aug 2021 - Jun 2022*
Bachelor's Thesis Project

- *Supervisor:* Prof. Babak Hossein Khalaj
- Built a full mobile network from open-source parts to put a controller above the radio: implemented 4G alongside standalone and non-standalone 5G deployments, then added an O-RAN RIC and a slicing xApp so that spectrum could be partitioned between tenants from a central controller, using OAI, FlexRAN/FlexRIC (Mosaic5G), and Open5GS.

Teaching Experience

CSE 260: Parallel Computation, Instructor: Bryan Chin *Fall 2026*
- Held discussion and Q&A sessions, and graded Exams

CSE 123: Computer Networks, Instructor: Prof. Schulman *Spring 2026*
- Held discussion and Q&A sessions, and graded Exams

Game Theory, Instructor: Prof. Ashtiani *Spring 2024*
- Designed problem sets, held Q&A sessions, and graded homework

Numerical Optimization, Instructor: Prof. Babaie-Zadeh *Fall 2023*
- Designed problem sets, held Q&A sessions, and graded homework

Probability and statistics, Instructor: Prof. Ashtiani *Fall 2023*
- Designed problem sets, held Q&A sessions, and graded homework of +20 students

Data Networks, Instructor: Prof. Pakravan *Fall 2022*
- Designed problem sets, held Q&A sessions, and graded homework of +30 students

Random Processes, Instructor: Prof. Behnia *Spring 2022*
- Held Q&A sessions, and graded homework of +20 students

Data Networks, Instructor: Prof. Pakravan *Spring 2022*
- Designed problem sets, held Q&A sessions, and graded homework of +30 students

Discrete-time Signal Processing, Instructor: Prof. Shamsollahi *Fall 2021*
- Designed problem sets and held Q&A sessions with +30 students

Circuits Theory, Instructor: Prof. Shamsollahi *Fall 2020*
- Designed problem sets and held Q&A sessions with +30 students

Electromagnetics, Instructor: Prof. Memarian *Spring 2020*
- Designed course website, designed problem sets, held Q&A sessions, and graded homework of +120 students

Honours and Awards

- **NSF Student Travel Grant**, IEEE DySPAN *2026*
- **NSF POSE Travel Grant**, Open6G+AI School *2026*
- **IEEE Eta Kappa Nu (IEEE-HKN) Honor Society**, UC San Diego *2026*
  Awarded to the top students in the graduate program.
- **Graduate Fellowship winner**, UC San Diego *2024*
- **Ranked 2nd** in my master's cohort, Electrical Engineering (Communication Systems) *Nov 2023*
- **Ranked 2nd** among EE BS graduates completing within the standard 4-year timeframe *2018 -2022*
- **Ranked 5th** overall out of 176 students in the EE BS graduating cohort *2018 -2022*
- **Exceptional Talents Award**, Sharif University of Technology *Feb 2022*
  Granted to the top 3% of students; confers direct admission to the MSc program, waiving the national entrance exam.
- **Silver Medal** in the 31st National Physics Olympiad *2017*
  Ranked among the top 40 nationally in a multi-stage competition open to all high school students in Iran; qualified for the IPhO Team Selection Camp.

- **Silver Medal**, International Mathematics Competition (IMC), Iran National Round *2013*
  Awarded to top performers in a nationwide mathematics competition open to all middle school students in Iran.

Presentations and Invited Talks

- **DySPAN**: Tiny-twin: A CPU-Native Full-stack Digital Twin for 5G *2026*
  IEEE International Symposium on Spectrum Innovation, Washington, DC, USA
- **OpenAirInterface workshop**: Tiny-twin: A CPU-Native Full-stack Digital Twin for 5G *2025*
  Hosted by UT Austin, openairinterface
- **MobiCom OpenAI workshop**: Log Pruning for Efficient Q&A in 5G Networks *2025*
  The 31st Annual International Conference on Mobile Computing and Networking Nov. 4-8, 2025, Hong Kong, China
- **6G Summit Abu Dhabi**: 5G Network Automation Using LLM Models and RAG *2024*
  Hosted by IIT, UAE
- **Ericsson AI/ML/6G Weekly**: LLM for 5G: Network Management *2024*
  Hosted by Amin Azari, Sweden
- **ICMLCN**: LLM for 5G: Network Management *2024*
  IEEE International Conference on Machine Learning for Communication and Networking, Sweden

Media Coverage

- **UC San Diego Today**: feature on Tiny-Twin *May 2026*
  "UC San Diego Team Launches Free Digital Twin for End-to-End Testing of Applications Over Wireless," first-authored work covered ahead of its IEEE DySPAN presentation — today.ucsd.edu

Selected Projects

Game Theory
DA matching Algorithm Implementation *Spring 2023*
- Assigning patients with special disease types to hospitals using Python. Each hospital has a disease preference and each disease has a hospital preference.

Numerical Optimization
Implementation of Optimization Algorithms *Fall 2022*
- Implementing line search based on strong Wolfe condition, steepest descent, Newton, and BFGS algorithms in MATLAB.

Massive Data Analysis
City Traffic Analysis *Fall 2022*
- Finding hot spots and rush hour of a city based on camera data of cars in a city.

Deep Learning
Sentiment Detection *Fall 2022*
- Fine-tuning multi-modal models and transformers to detect sentiment based on text and image.

Queuing Theory
Simulation of M/M/m queue *Fall 2022*
- Simulating different Queuing scenarios including queues that contain positive and negative signals using Python.

Cloud Software Mobile Network
Routing Algorithm Implementation *Spring 2022*
- Implementing dynamic routing with the Dijkstra algorithm on an SDN controller(ODL) in Python.

Advanced Data Networks
MIPS Processor Implementation *Spring 2022*
- Implementing Multi-Cycle and Single Cycle architecture of MIPS Processor using Verilog.

Information Theory
Distributed Coding *Fall 2021*
- Presenting and providing a term paper on distributed coding.

Data Networks
Simplified RPC Protocol *Fall 2021*
- Implementing a simplified RPC protocol using socket programming and multi-threading in Python.

Machine Learning
Biometric Authentication *Spring 2021*
- Biometric Authentication using ECG signals. Preparing noisy signals with MATLAB filtering toolbox and using

SVM for classification.

Convex Optimization

Convex Optimization in Economics *Spring 2021*

- Convex optimization on an economic problem using MATLAB and CVX.

Computer and Microprocessor architecture

MIPS Processor Implementation *Spring 2020*

- Implemented Multi-Cycle and Single Cycle architecture of MIPS Processor using Verilog.

Programming Skills

MATLAB & Simulink

- Proficient in MATLAB programming (signal processing, CVX, digital filter design, and implementing machine learning algorithms and neural networks)
- Proficient in using Simulink to simulate communication systems and control systems.

Python Programming

- Proficient in programming using Python and experienced in working with libraries (CvxPy, sklearn, SciPy, TensorFlow, PyTorch, Spark, Pandas) and implementing learning algorithms using Python.

Other programming and tools: C/C++, go, bash, Kubernetes, ODL, NS3, GNS3, Git, Verilog, Assembly Mininet, Linux, Docker, CUDA, Claude Code, Codex, NotebookLM

Selected Coursework

University of California San Diego

- Information Theory (A+)
- Modern Communication Networks(A+)
- Principles of Wireless Networks(A)
- Linear Algebra and Application(A)
- Computer Security(A+)
- Networked Services(A+)
- Parallel Computation(A)
- Machine Learning for Physical Applications(A)

Sharif University of Technology

- Queuing Theory(20/20 - Graduate)
- Advanced Data Networks(19.5/20 - Graduate)
- Game Theory(20/20 - Graduate)
- Information Theory(19.1/20 - Graduate)
- Numerical Optimization(19.8/20 - Graduate)
- Data Networks(19.5/20 - Graduate)
- Random Processes(20/20- Graduate)
- Digital Communication(20/20)
- Convex Optimization(19.5/20)
- Communication Systems(19.6/20)
- Digital Signal Processing(20/20)
- Linear Control Systems(20/20)
- Introduction to Machine Learning(20/20)
- Probability and Statistics(19.7/20)

Professional Service

- **Reviewer**, IEEE Transaction on Computers *2026*
- **TPC Member and Reviewer**, ACM S3 Workshop (MobiCom) *2026*
  Workshop on Wireless of the Students, by the Students, and for the Students, co-located with ACM MobiCom
- **Reviewer**, IEEE Transactions on Dependable and Secure Computing *2026*
- **Reviewer**, IEEE Access *2026*
- **Reviewer**, IEEE Future Networks World Forum *2026*
- **Reviewer**, JMIR Formative Research *2026*
- **Reviewer**, IEEE ICASSP *2027*
- **Student Volunteer**, IEEE DySPAN *2026*
- **Student Volunteer**, ACL *2026*
  Annual Meeting of the Association for Computational Linguistics

Mentorship

- **University of California San Diego**
  Qingyuan Zheng (now a Ph.D. student at Duke University), Wooil Jung, Nikhil Kuniyil: mentored on 5G log analysis and protocol state machine recovery; co-authored resulting publications.
- **Sharif University of Technology**
  Alireza Shokrani, Seyed Mohsen Nasiri, Sepehr Kazemi Ranjbar, Parsa Hatami, Ahmadreza Majlesara: mentored on

LLM-based 5G network automation; co-authored resulting publications.

Language Skills

Persian: Native proficiency

English: Professional proficiency

Spanish: Elementary Proficiency